

• Disadvantages of Half wave Rectifier.

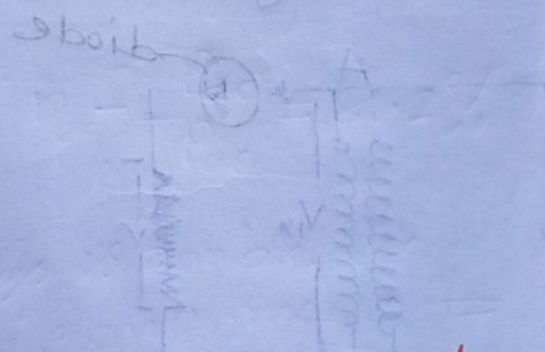
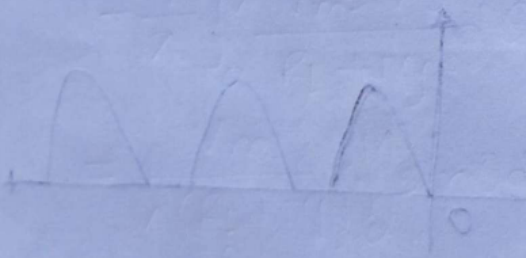
(i) The pulsating current in the load contains alternating component whose basic frequency is equal to the supply frequency ($f_{out} = f_{in}$). Therefore, an elaborate filtering is required to produce a steady direct current.

(ii) The a.c. supply delivers power only half the time. Therefore, the output is low.

• Efficiency of ^{Half} full wave Rectifier $\rightarrow AC \rightarrow DC$

(b) Output $\left[\eta = \frac{\text{DC output power}}{\text{input ac power}} \right]$ Input (ac)

Diode conducts and supplies power to the load only during positive half cycle of the input ac supply.



Output dc power is pulsating dc. So for output dc power we need to find the average load current.

Efficiency of Half wave Rectifier.

Let, voltage at the secondary side of the transformer $V = V_m \sin \theta$,

Resistances of the diode and the load are r_f and R_L . So the average current,

$$I_{av} = \frac{1}{2\pi} \int_0^\pi \frac{V}{R} d\theta$$

$$= \frac{1}{2\pi} \int_0^\pi \frac{V_m \sin \theta}{R_L + r_f} d\theta$$

$$= \frac{V_m}{2\pi (R_L + r_f)} \int_0^\pi \sin \theta d\theta$$

Forward diode as a resistor

$$= \frac{V_m}{2\pi (R_L + r_f)} * 2$$

here, $V_m = I_m (R_L + r_f)$

$$= \frac{V_m}{\pi (R_L + r_f)} = \frac{I_m}{\pi}$$

• Output dc power $P_{dc} = I_{av}^2 * R_L = \left(\frac{I_m}{\pi}\right)^2 * R_L$

• Input ac power $P_{ac} = I_{rms}^2 * (R_L + r_f)$
 $= \left(\frac{I_m}{2}\right)^2 * (R_L + r_f)$

(here, $I_{rms}(\text{for half cycle}) = \sqrt{\frac{1}{2} \times \frac{1}{2\pi} \int_0^{2\pi} \left\{ \frac{V_m \sin \theta}{(R_L + r_f)} \right\}^2 d\theta}$)

$$= \frac{I_m}{2}$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{(I_m/r_f)^2 * R_L}{(I_m/2)^2 * (R_L + r_f)}$$

$$= \frac{0.406 R_L}{R_L + r_f} = \frac{0.406}{1 + \frac{r_f}{R_L}} = 0.406 \times 100\% = 40.6\%$$

$r_f \ll R_L$, so, $1 + \frac{r_f}{R_L} \approx 1$

diode এর মধ্যে থাকা resistance কে বড় মান দ্বারা ভাগ করলে ০ এর কাছাকাছি মান আসবে যা বর্জনযোগ্য

Max rectifier efficiency = 40.6%.

- sin wave 2A rms value $(\frac{I_m}{\sqrt{2}})$ Full wave
- sin wave A rms value $(\frac{I_m}{2})$ Half wave

Efficiency of Full wave Rectifier.

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \left\{ \frac{V_m \sin \theta}{R_L + r_f} \right\}^2 d\theta} = \frac{I_m}{\sqrt{2}}$$

$$I_{av} = \frac{1}{2\pi} \int_0^{2\pi} \frac{V}{R} \sin \theta d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{V_m \sin \theta}{R_L + r_f} d\theta$$

$$= \frac{V_m}{2\pi(R_L + r_f)} \int_0^{2\pi} \sin \theta d\theta$$

$$I_{av} = \frac{1}{\pi} \int_0^{\pi} \frac{V}{R} d\theta$$

$$= \frac{1}{\pi} \int_0^{\pi} \frac{V_m \sin \theta}{R_L + r_f} d\theta$$

$$= \frac{V_m}{\pi(R_L + r_f)} \int_0^{\pi} \sin \theta d\theta$$

$$= \frac{V_m}{\pi(R_L + r_f)} * 2$$

$$= \frac{2V_m}{\pi(R_L + r_f)} = \frac{2I_m}{\pi}$$

$$P_{ac} = I_{rms}^2 * (R_L + r_f)$$

$$= \left(\frac{I_m}{\sqrt{2}} \right)^2 * (R_L + r_f)$$

$$\eta = \frac{P_{dc}}{P_{ac}} = 81.2\%$$

This is double the efficiency due to half wave rectifier.

Therefore, a full wave rectifier is twice as effective as a half wave rectifier.

Full wave Rectifier

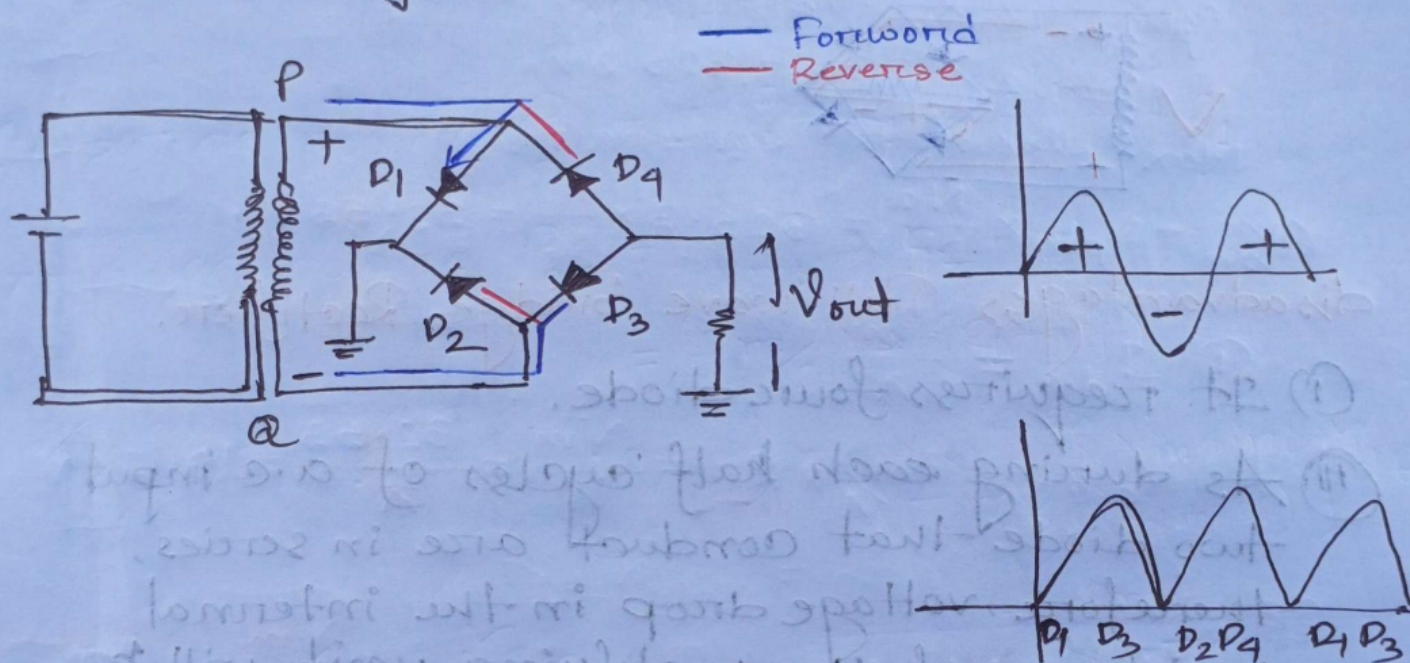
The following two circuits are commonly used for full wave rectification.

- (i) Centre-tap full wave rectifier.
- (ii) Full wave bridge rectifier.

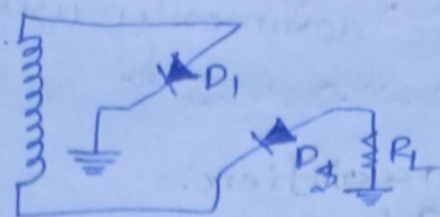
Working.

During positive half cycle diode D_1 and D_3 are forward biased, D_2 and D_4 are reverse biased.

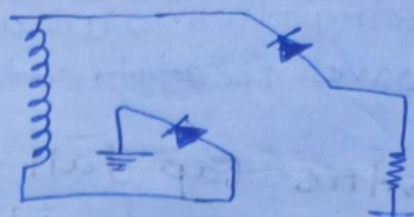
So current flows through D_1, R_L, D_3 and return back to the secondary winding of the transformer. output voltage is obtained across the load R_L .



Load current path for positive and negative cycle of the input ac.



(i)

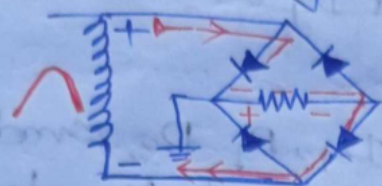


(ii)

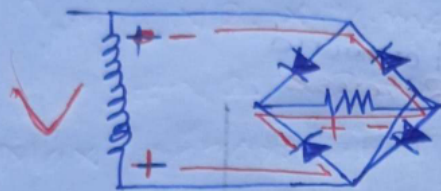
Here $I_{out} = 2I_{in}$

● Full wave bridge Rectifier.

Positive half cycle



Negative half cycle



Disadvantages Full wave bridge Rectifier.

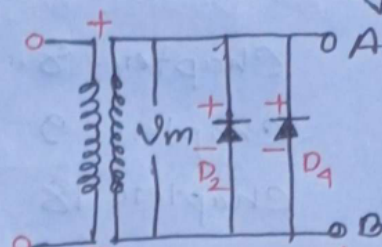
(i) It requires four diode.

(ii) As during each half cycles of a.c input two diode that conduct are in series, therefore, voltage drop in the internal resistance of the rectifying unit will be twice as great as in the center tap circuit. This is objectionable when secondary voltage is small.

● Peak Inverse Voltage.

is the maximum voltage that can be applied across the diode during reverse biased condition. It must be less than the breakdown voltage of the diode.

Depends of circuit

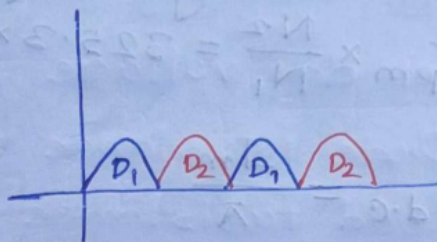
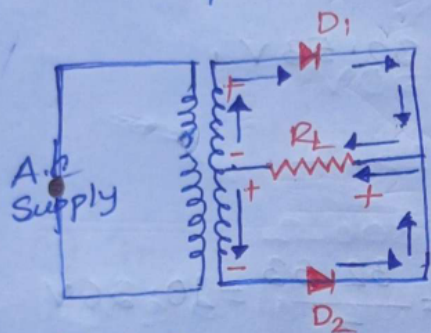


Here the PIV of the diode is V_m volt.

Topics	Half wave	Full wave
I_{avg}	I_m / π	$2I_m / \pi$
I_{rms}	$I_m / 2$	$I_m / \sqrt{2}$
Efficiency	40.6	81.2

Online + 02.

● Center tap full wave rectifier



Peak Inverse Voltage = $2V_m$

Disadvantage:

- It is difficult to locate the center tap on the secondary winding.
- The d.c output is small as each diode utilise only half of the transformer secondary voltage.
- The diodes used must have high peak inverse voltage.

Principles of electronics - Chand - V - S mehta

rohit mehta.

Chapter - 6 6.13-6.21, 6.23, 6.24

*** 6.14, 6.16, 6.18, 6.19, 6.20 ***

Chapter - 8. 8.1-8.12, 8.23, 8.25, 8.32-8.34,

Chapter - 9 9.21, 9.22

Chapter 18 18.5-18.8, 18.10, 18.12, 18.13,
18.14, 18.18-18.20.

Example 6.13.

Primary to secondary turns is $\frac{N_1}{N_2} = 10$.

R.M.S primary voltage = 230V.

∴ Max primary voltage is,

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_{pm} = \sqrt{2} \times \text{r.m.s primary voltage} \\ = \sqrt{2} \times 230 = 325.3V$$

Max secondary voltage is,

$$E_{sm} = V_{pm} \times \frac{N_2}{N_1} = 325.3 \times \frac{1}{10} = 32.53V$$

$$\textcircled{1} I_{d.c} = \frac{I_m}{\pi}$$

$$V_{d.c} = I_{m/\pi} \times R_L = \frac{V_m}{\pi} = \frac{32.53}{\pi} = 10.36V$$

230/sinwt

→ Peak value.

230 → r.m.s

(i) During the negative half cycle of a.c supply the diode is reverse biased and hence conducts no current. Therefore the maximum secondary voltage appears across the diode
 $\therefore$ Peak inverse voltage 32.53 V

a.c - rms

d.c - avg.

Ex 6.14 A crystal diode, having internal resistance $r_f = 20 \Omega$ is used for half-wave rectification. If the applied voltage $V = 50 \sin \omega t$ and load resistance $R_L = 800 \Omega$ find.

(i) I_m, I_{dc}, I_{rms}

(ii) a.c power input and d.c power output.

(iii) d.c output voltage.

(iv) efficiency of rectification.

$$V = 50 \sin \omega t$$

$$\text{Maximum voltage } 50 \text{ V} = V_m$$

$$(i) I_m = \frac{V_m}{r_f + R_L} = \frac{50}{20 + 800} = 0.061 \text{ A} = 61 \text{ mA}$$

$$I_{dc} = \frac{I_m}{\pi} = \frac{61}{\pi} = 19.4 \text{ mA}$$

$$I_{rms} = I_m / 2 = 61 / 2 = 30.5 \text{ mA}$$

$$\begin{aligned}
 \text{(ii) ac power input} &= (I_{\text{rms}})^2 \times (r_f + R_L) \\
 &= \left(\frac{30.9}{1000}\right)^2 \times (20 + 800) \\
 &= 0.763 \text{ watt}
 \end{aligned}$$

$$\begin{aligned}
 \text{dc power output } I_{\text{dc}} \times R_L &= \left(\frac{19.4}{1000}\right) \times 800 \\
 &= 0.301 \text{ watt}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) dc output voltage } I_{\text{dc}} R_L \\
 &= 19.4 \times 800 = 15.32 \text{ volts.}
 \end{aligned}$$

$$\text{(iv) Efficiency} = \frac{0.301}{0.763} \times 100 = 39.5\%$$

Example 6.15. ***

A half wave rectifier is used to supply 50V dc. to a resistance load of 800Ω . The diode has a resistance of 25Ω . Calculate ac require voltage. (Asking for supply)

Output dc voltage $V_{\text{dc}} = 50 \text{ V}$

Diode resistance $r_f = 25 \Omega$

Load resistance $R_L = 800 \Omega$

Let V_m be the maximum value of a.c voltage required.

$$\begin{aligned}
 V_{\text{dc}} &= I_{\text{dc}} \times R_L \\
 &= \frac{I_m}{\pi} \times R_L = \frac{V_m}{\pi(r_f + R_L)} \times R_L
 \end{aligned}$$

$$V_m = \frac{\pi \times 825 \times 30}{800}$$

$$= 162 \text{ V}$$

$$I_m = \frac{V_m}{r_f + R_L}$$

Formula

6.14 A full wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer r.m.s secondary voltage from centre tap to each end of secondary is 50 V and load resistance is $980\ \Omega$.

Find: (i) the mean load current

(ii) the r.m.s value of load current.

Solution.

$$r_f = 20\ \Omega, R_L = 980\ \Omega$$

Each End

$$\text{Max. ac. voltage } V_m = 50 \times \sqrt{2} = 70.7\text{ V}$$

$$\text{Max load current } I_m = \frac{V_m}{r_f + R_L} = \frac{70.7}{(20 + 980)\ \Omega} = 70.7\text{ mA}$$

Secondary
Voltage
Each End
50/2

(i) Mean load current

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 70.7}{\pi} = 45\text{ mA}$$

(ii) R.M.S value of load current is

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{70.7}{\sqrt{2}} = 50\text{ mA}$$